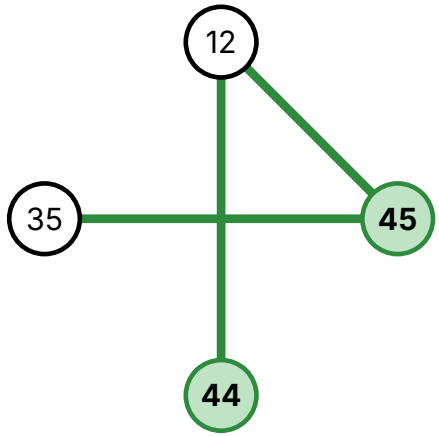
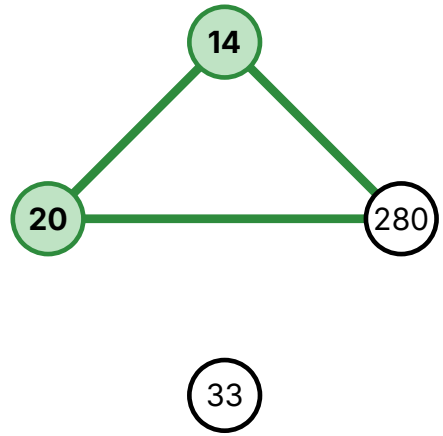
$D = [12, 35]$: $f = (2,)$, $\beta = (2)$

$D \cup N$, $N = [44, 45]$: $f = (4, 3)$, $\beta = (1)$



$D = [33, 280]$: $f = (2,)$, $\beta = (2)$

$D \cup N$, $N = [14, 20]$: $f = (4, 3)$, $\beta = (2, 1)$



both columns: $f = (2, 0, 0) \to (4, 3, 0)$, nothing lost, 2 new vertices and 3 new edges; $\Delta\beta = (-1, 0, 0)$ versus $(0, +1, 0)$ (SIV-P7)